Ridgeline Weather Network

Data-health assessment and field recommendation | Spring 2025
Observation window: 1 March–11 April 2025 (UTC)

Decision at a glance

The network is usable for a spring climate summary after targeted quality control. Retain valid observations from all eight stations rather than rejecting whole stations because of their total flag counts. Exclude demonstrably invalid values, resolve duplicate timestamps before calculating affected statistics, and document gaps and unresolved calibration concerns.

The records cover the same broad early-spring period, so timestamp coverage alone does not determine data quality. The decision should be made by channel and affected time interval: a station may have useful temperature data while its humidity, pressure, or timing records need separate treatment.

Findings and treatment

Evidence	Assessment	Recommended treatment
S05 — relative humidity above 100%	Readings reported at 104% are outside the physically valid range for relative humidity.	**Exclude** the above-100% humidity readings from humidity summaries. Keep other channels and valid periods; do not discard the whole station.
S02 — pressure board replaced	The maintenance note says the replacement board was not calibrated. Plausible pressure values can still carry a systematic offset.	Use pressure cautiously and mark it provisional until checked against a trusted reference. Do not infer that the entire station is bad.
S06 and S08 — duplicate timestamps	Duplicate times may represent repeated identical records or conflicting measurements; timestamps alone cannot distinguish them.	Compare the records channel by channel. Deduplicate identical copies; investigate conflicting values before deciding which record, if any, to exclude. Do not automatically discard every duplicate.
S08 — missing interval	A gap of roughly 35 hours 50 minutes interrupts the otherwise regular 10-minute sequence.	Report the coverage gap and do not treat interpolated values as observations. Use the remaining valid data.
GPS logs — invalid checksums / void fixes	These records are not reliable for GPS-dependent use.	Exclude invalid GPS records from location/timing checks. This does not by itself invalidate the station's weather measurements.
Other station records	Station-level flag totals are useful for triage, but a lower total does not prove that every channel is reliable.	Retain observations that pass the relevant checks, with exclusions and limitations recorded at the channel/time-interval level.

Which data is good enough for the spring climate summary?

Use valid observations from all eight stations for the relevant climate statistics after applying the checks above. Exclude S05 relative-humidity values above 100%; resolve S06/S08 duplicate timestamps before using affected records; and report S08's missing interval as a coverage limitation. Treat S02 pressure as provisional until calibration is verified. Invalid GPS records should be excluded from GPS-dependent analyses, not used as a reason to discard unrelated weather channels.

Calculate each statistic only from records that pass the checks for that channel, and state the effective sample coverage after exclusions. Do not silently fill missing periods or present uncertain pressure measurements as fully verified.

Where should the technician go next week?

First: S02. Spend the hour checking the pressure channel against a trusted, recently calibrated barometer at the same time and location. Record both readings, confirm units and sensor configuration, inspect wiring and the replacement board, and check whether the offset is stable. Do not apply a correction unless the comparison supports a defensible correction method.

Why S02 over S05? S02's board replacement without calibration creates uncertainty that may affect an entire run of otherwise plausible pressure observations. S05's above-100% humidity values are already identifiable and can be excluded from humidity calculations while its other data remain usable. S05 is the next priority for a focused humidity-sensor inspection.

This assessment separates confirmed invalid values from records that need verification. Flag counts are a triage aid, not a standalone station-quality score; final inclusion depends on the channel, time interval, and evidence available.